PROBABILITY DICE & MODULAR SPINNER PACK

Random Variables, Sample Spaces & Statistical Distributions

CARD 07 / 10 • PROB-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · 6·Algebra · **7·Probability** · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

SAMPLE SPACES, 2d6 SUMS & NORMAL CURVES

Randomness follows strict combinatorial laws.
Rolling two fair 6-sided dice gives 36 outcomes:
Sample Space = 6 × 6 = 36 Pair Outcomes
2d6 sum forms a triangle distribution peak at 7!

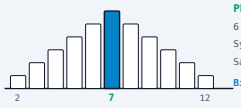
CALIBRATED 16.00 MM FAIR DICE DATUM

- Fair d6: 16.00 mm precision cube, balanced pips
- Polyhedral d20: 20-sided icosahedron (1/20 odds)
- Low-Friction Spinner: Precision brass bushing pivot
- Dials: 10-sector uniform & binary coin-flip

2d6 SAMPLE SPACE GRID & SPINNER POST

Lid features 6×6 outcome matrix & spinner post.
Watch experimental trials converge to exact odds!

2d6 SUM PROBABILITY: TRIANGULAR PEAK AT 7 (6/36)



PEAK AT 7 (16.7%)
6 / 36 Combinations
Symmetric Triangle
Sample Space: 6×6 = 36
Birth of the Bell Curve

2. KIT PARTS INVENTORY (14 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Precision Fair d6	4x	16.0 mm Volume-Balanced	High-Contrast
Polyhedral d20 Die	1x	20-Sided Icosahedron	Emerald Green
Low-Friction Spinner	1x	Brass Pivot Precision Hub	High-Spin Brass
Interchangeable Dials	4x	10-Sector / Binary / Custom	Polymer Plates
2d6 Sample Space Lid	1x	36-Outcome Matrix Grid	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 10 Working Parts + 2 Extra Dials + Storage Bin + Lid = 14 Parts

PROBABILITY DISTRIBUTION PALETTE

- 4 Color-Separated d6 Dice: Enables independent event tracking.
- Emerald d20 Die: High-visibility prime distribution testing.
- High-Spin Brass Bushing: Ultra-low friction for random spins.
- Clear Calibration: High-contrast legible numerals.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Seat the 4 d6 dice and 1 d20 into their molded bottom bays. Stack the spinner dials flat and click the brass spinner hub onto its retention post. Any empty bay shows. Lid latches firmly for silent, rattle-free carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

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CENTRAL LIMIT THEOREM & LAW OF LARGE NUMBERS

The sum of independent random variables tends toward a normal distribution as sample size grows.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

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SERIES 1 • CARD 07 OF 10 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS • 2d6 36-OUTCOME MATRIX • WEIGHTED HARMONIC SPINNER

LESSON GUIDE: 2d6 SAMPLE SPACES & INDEPENDENT EVENTS

Step-by-Step Hands-On Tutorial: Why 7 is the Most Likely Roll (P = 6/36)

Grades 5–10 Probability & Statistics • Hands-On Classroom & Home STEM Studio Series

CARD 07 / 10 • PROB-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/dice-and-spinner/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.7.SP.C.8 (Compound Events)
- CCSS.MATH.CONTENT.HSS.CP.A.2 (Independence)
- TEKS 111.27.b.6.B (Representations)

Grades 5–10 Probability & Statistics

MATHEMATICAL INQUIRY

Why are there six different combinations to roll a 7, but only one combination to roll a 2?

1 STEP 1: INSPECT THE 6 × 6 SAMPLE SPACE

Examine the 6 × 6 grid on the lid stage.
Die 1 rows (1–6) × Die 2 columns (1–6) = 36 total outcomes.
Every individual pair (like [3,4] or [6,1]) has probability 1/36.

2 STEP 2: COUNT PATHS TO EACH SUM

Trace diagonals of constant sum across the grid:

- Sum 2: Only 1 way [1,1] → P = 1/36 (2.8%)
- Sum 7: 6 ways [1,6],[2,5],[3,4],[4,3],[5,2],[6,1] → P = 6/36 (16.7%)
- Sum 12: Only 1 way [6,6] → P = 1/36 (2.8%)

3–4 STEPS 3 & 4: SAMPLE TRIALS & CALCULATE

3. SAMPLE EXPERIMENTAL RUNS: Roll 2d6 three times: 7, 4, 10.

4. COMPUTE SAMPLE MEAN: Calculate mean on calculator:

Press Dice & Statistics Icon ([🎲], [Σ+] & [g][x̄]) and run [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → 7.

Independent Event: d6=6 and Spinner=10 → 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → 1/60.

5 CORE PRINCIPLE: THE BIRTH OF THE BELL CURVE

Single dice have flat uniform distributions (each face is 1/6). But when two dice add together, extreme sums require extreme coordination, while central sums have many different additive combinations! This combinatorial clustering is the foundation of the normal distribution Bell Curve!

★ EXTENSION INVESTIGATION: EVEN SUM PROBABILITY

Count all 2d6 combinations that give an EVEN sum (2, 4, 6, 8, 10, 12). Exactly 18 out of 36 outcomes! P = 50.0%! (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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CROSS-REFERENCED CALCULATOR VERIFICATION

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